Assignment - 1

Name: Sanket Rath
PRN: 202301100006

Topic: Analysis of a real-world application's DevOps lifecycle and design of its workflow/architecture. Implement basic configuration management using Ansible with an inventory, playbook, screenshots, and a short demo.

Steps:

1. Choosing a Real World Scenario:

   Scenario: Production E-Commerce Platform (Two-Tier Architecture)

   Web Tier: Nginx Reverse Proxy serving dynamic frontend traffic and routing requests to an API.
   Backend/Data Tier: Python/Node backend backed by a relational database (PostgreSQL / MySQL).

2. DevOps Lifecycle Breakdown

   - **Plan and Code:** Developers use Git and GitHub to collaborate and work together. The Repository could have multiple branches:
     - Main branch for production
     - Develop for staging etc

- **Build and test:** This is the part of continuous integration. Every time the code is pushed on the main branch, the automated tests run on the code.

  It performs unit testing to make sure there are no broken logic in the code.

- **Deployment:** This is where Ansible comes in. Ansible acts as an automated installer and setup assistant for your servers.

  Instead of an engineer manually logging into each computer via a terminal to type commands one by one, Ansible does the entire setup automatically using a single script called a Playbook.

- **Release:** The server starts running, making the website live for customers.

- **Monitoring:** Tools watch the server in background to ensure it has enough memory and does not crash.

3. Understanding Ansible's Working

Ansible automates server configuration through an **agentless, push-based model**:

- **Reads Configuration:** On your main computer (the *Control Node*), Ansible reads your target list and the instructions you wrote.
- **Connects via SSH:** It establishes a standard, secure SSH connection to the remote target machines (*Managed Nodes*). No background agents or daemons need to be installed on target servers beforehand.
- **Executes Modules:** It translates your instructions into small Python scripts (called *Modules*), sends them over SSH to the target machine, and executes them.
- **Applies Changes & Cleans Up:** The module verifies the system's state, applies any needed changes, removes the temporary scripts, and reports back whether tasks were `ok` (already correct), `changed` (updated), or `failed`.

Key files needed in Ansible: To run any automation you need the following two fundamental files:

1. Inventory File (inventory.ini or hosts):
   - **What it is:** The target address book.
   - **What it does:**
     - Defines the IP addresses or domain names of all servers you want to configure.
     - Organizes servers into logical groups (e.g., `[webservers]`, `[dbservers]`).
     - Specifies connection settings, such as SSH usernames, private keys, or whether to run locally (`ansible_connection=local`).

2. Playbook File (playbook.yml):
    - **What it is:** The configuration recipe written in YAML format.
    - **What it does:**
        - Maps specific tasks to target groups defined in the inventory.
        - Declares the **desired end state** of the target machine (e.g., "Nginx must be installed", "file `index.html` must exist", "service must be active").
        - Ensures **idempotency**: if a setting or package already matches the instructions, Ansible skips it without making redundant changes or breaking existing setups.

4. Using KillerCoda for installing Ansible and creating the required files.

```
sanket@sanket-VirtualBox:~$ sudo apt update && sudo apt install -y ansible
[sudo] password for sanket:
Get:1 https://dl.google.com/linux/chrome/deb stable InRelease [2,548 B]
Hit:2 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 https://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1,412 B]
Get:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,144 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,397 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [308 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu noble-updates/main Icons (48x48) [37.3 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu noble-updates/main Icons (64x64) [52.7 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.7 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1,805 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [230 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/main Icons (48x48) [19.2 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/main Icons (64x64) [26.7 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [411 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11.9 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1,713 kB]
```

Confirming the installation:

```
sanket@sanket-VirtualBox:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/sanket/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/sanket/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 18 2025, 17:59:45) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
sanket@sanket-VirtualBox:~$
```

Creating a directory

```
sanket@sanket-VirtualBox:~$ mkdir -p ~/ecommerce-ansible && cd ~/ecommerce-ansible
sanket@sanket-VirtualBox:~/ecommerce-ansible$
```

Create the inventory.ini file

```
  GNU nano 7.2
[webservers]
web_node_01 ansible_host=localhost ansible_connection=local

[dbservers]
db_node_01 ansible_host=localhost ansible_connection=local

[ecommmerce:children]
webservers
dbservers

[all:vars]
ansible_python_interpreter=/usr/bin/python3
```

Test connectivity for inventory hosts

```
sanket@sanket-VirtualBox:~/ecommerce-ansible$ ansible -i inventory.ini all -m ping
web_node_01 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
db_node_01 | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
sanket@sanket-VirtualBox:~/ecommerce-ansible$
```

Creating the playbook.yml file

```
  GNU nano 7.2                                                    playboo
---
- name: Provision E-Commerce Storefront Web Tier
  hosts: webservers
  become: true

  vars:
    app_title: "Cloud-Native E-Commerce Storefront"
    environment_name: "Production"

  tasks:
    - name: 1. Update system package repository
      apt:
        update_cache: yes
        cache_valid_time: 3600

    - name: 2. Install Nginx web server and utilities
      apt:
        name:
          - nginx
          - curl
        state: present

    - name: 3. Deploy E-Commerce storefront landing page
      copy:
        dest: /var/www/html/index.html
        owner: www-data
        group: www-data
        mode: '0644'
        content: |
          <!DOCTYPE html>
          <html>
          <head>
            <title>{{ app_title }}</title>
            <style>

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

5. First Run of the Playbook

```
sanket@sanket-VirtualBox:~/ecommerce-ansible$ sudo ansible-playbook -i inventory.ini playbook.yml

PLAY [Provision E-Commerce Storefront Web Tier] ********************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [web_node_01]

TASK [1. Update system package repository] *************************************************************************
ok: [web_node_01]

TASK [2. Install Nginx web server and utilities] *******************************************************************
ok: [web_node_01]

TASK [3. Deploy E-Commerce storefront landing page] ****************************************************************
changed: [web_node_01]

TASK [4. Ensure Nginx service is running and enabled] **************************************************************
ok: [web_node_01]

PLAY RECAP *********************************************************************************************************
web_node_01                : ok=5    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

sanket@sanket-VirtualBox:~/ecommerce-ansible$
```

6. Verifying the Deployment of Website

```
sanket@sanket-VirtualBox:~/ecommerce-ansible$ curl -i http://localhost
HTTP/1.1 200 OK
Server: nginx/1.24.0 (Ubuntu)
Date: Tue, 13 Oct 2026 12:13:05 GMT
Content-Type: text/html
Content-Length: 1033
Last-Modified: Tue, 13 Oct 2026 12:10:46 GMT
Connection: keep-alive
ETag: "6ace1fc6-409"
Accept-Ranges: bytes

<!DOCTYPE html>
<html>
<head>
  <title>Cloud-Native E-Commerce Storefront</title>
  <style>
    body { font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, san
    .container { max-width: 650px; margin: auto; background: #1e293b; padding: 30p
    h1 { color: #38bdf8; margin-bottom: 8px; }
    .badge { display: inline-block; padding: 4px 12px; background: #0369a1; border
    p { color: #94a3b8; line-height: 1.6; }
    .status { color: #4ade80; font-weight: bold; }
  </style>
</head>
<body>
  <div class="container">
    <span class="badge">Production Environment</span>
    <h1>🛒 Cloud-Native E-Commerce Storefront</h1>
    <p>Deployment Status: <span class="status">Active & Idempotent</span></p>
    <p>Automated via <strong>Ansible Configuration Management</strong></p>
  </div>
</body>
</html>
sanket@sanket-VirtualBox:~/ecommerce-ansible$
```

7. Verifying on the Browser

